

Loïc Labache, Ph.D.

Postdoctoral Associate in Psychiatry

Department of Psychiatry,
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ORCID ▪ Scholar

Academic Appointments

- 2023 – Present **Postdoctoral Associate**, *Department of Psychiatry*, Rutgers University – Piscataway, NJ, US. PI: Prof. Avram J. Holmes
- 2023 – 2025 **Research Collaborator**, *Department of Psychology*, Yale University – New Haven, CT, US. PI: Prof. Gregory McCarthy
- 2021 – 2023 **Postdoctoral Associate**, *Department of Psychology*, Yale University – New Haven, CT, US. PI: Prof. Avram J. Holmes
- 2020 – 2021 **Postdoctoral Associate**, *Department of Epidemiology, French National Institute of Health and Medical Research (Inserm)*, Université de Bordeaux – Bordeaux, FR. PI: Dr. Cécilia Samieri

Education

- 2017 – 2020 **Ph.D.** (Neuroimaging & Cognitive Neuroscience), *Doctoral School of Mathematics and Computer Science – Université de Bordeaux*, Bordeaux, FR
Dissertation: Elaboration of Brain Network Atlases Underpinning Lateralized Cognitive Functions
PI: Dr. Nathalie Tzourio-Mazoyer, Dr. Marc Joliot, & Prof. Jérôme Saracco
- 2014 – 2017 **M.S.** (Engineering Degree), *Institut Polytechnique de Bordeaux*, Talence, FR
Computer Science, Cognitive Neurosciences, Applied Mathematics.
- 2011 – 2014 **B.S.**, *Université de Bordeaux*, Bordeaux, FR
Cognitive Neurosciences, Computer Science, Applied Mathematics.

Funding and Fellowships

Funded

- 2025 – 2026 **Postdoctoral Health AI Research Fellows**, *Rutgers Health Center for Biomedical Informatics & Health Artificial Intelligence*, Rutgers University – New Brunswick, US-NJ
- 2017 – 2020 **CEA Graduate Research Fellowship**, *French Alternative Energies and Atomic Energy Commission – Paris*, FR.

Pending Applications

- Pending **National Institute of Mental Health**, *NIH K99/R00 – Pathway to Independence Award*

Honors & Awards

- 2026 **Rutgers Brain Health Institute Trainee Travel Award**, *Brain Health Institute*, Rutgers University – Piscataway, US-NJ.
- 2024 **Aperture Neuro 2023 Best Paper Award**, *Organization for Human Brain Mapping – Seoul*, KR.
- 2023 **Runner-Up Trainee Prize**, *The Neuro - Irv and Helga Cooper Foundation Open Science Prizes – Montréal*, CA.
- 2023 **Merit Award Abstract Winner**, *Organization for Human Brain Mapping – Montréal*, CA.

Publications

Source codes and atlases are available on GitHub: github.com/loiclabache 

Peer-reviewed

18. **Labache, L.**, Chopra, S., Zhang, X.-H., & Holmes, A. J. (2026). Molecular and cellular correlates of human cortical lateralization. *Communications Biology*. DOI: 10.1038/s42003-026-10508-1 [Code]
17. Cocuzza, C. V., Chopra, S., Segal, A., **Labache, L.**, Chin, R., Joss, K., Holmes, A. J. (2026). Brain network dynamics reflect psychiatric illness status and transdiagnostic symptom profiles across health and disease. *Nature Communications*. DOI: 10.1038/s41467-026-75585-6 [Code]
16. Chopra, S., Worhunsky, P. D., Naganawa, M., Zhang, X.-H., Segal, A., **Labache, L.**, Orchard, E., Cropley, V., Wood, S., Angarita, G. A., Cosgrove, K., Matuskey, D., Nabulsi, N.B., Huang, Y., Carson, R. E., Esterlis, I., Skosnik, P. D., D'Souza, D. C., Holmes, A. J., Radhakrishnan, R. (2026). Widespread Synaptic Density Loss in Schizophrenia Follows Molecular and Network Architecture. *Molecular Psychiatry*. DOI: 10.1038/s41380-026-03717-x
15. Qu, Y. L., Chopra, S., Qu, S., Cocuzza, C. V., **Labache, L.**, Bauer, C. C. C., Morfini, F., Whitfield-Gabrieli, S., Slavich, G. M., Joormann, J., Holmes, A. J. (2026). Life experiences of humiliation, entrapment, and frontoparietal-cerebellar connectivity predict adolescent anxiety and depression symptoms. *Psychological Medicine*. DOI: 10.1017/S0033291726103699
14. **Labache, L.***, Roger, E.*, Hamlin, N., Kruse, J., Baci, M., & Doucet, G. E. (2025). When age tips the balance: a dual mechanism affecting hemispheric specialization for language. *Imaging Neuroscience*. DOI: 10.1162/IMAG.a.63 [Code] *authors contributed equally to this work
13. Chopra, S., Cocuzza, C. V., Lawhead, C., Ricard, J. A., **Labache, L.**, Patrick, L. M., Kumar, P., Rubenstein, A., Moses, J., Chen, L., Blankenbaker, C., Gillis, B., Germine, L. T., Harpaz-Rote, I., Yeo, B.T. T., Baker, J. T., Holmes, A. J. (2025). The Transdiagnostic Connectome Project: a richly phenotyped open dataset for advancing the study of brain-behavior relationships in psychiatry. *Scientific Data*. DOI: 10.1038/s41597-025-04895-z [Code]
12. Lawhead, C., Silver, J., Olino, T., **Labache, L.**, Juhng, S., Schwartz, H., Klein, D. (2025). Longitudinal clustering of psychopathology across childhood and adolescence: an approach toward developmentally-based classification. *Clinical Psychological Science*. DOI: 10.1177/21677026251357589
11. Ricard, J.A., **Labache, L.**, Segal, A., Dhamala, E., Cocuzza, C. v., Jones, G., Yip, S., Chopra, S., & Holmes, A. J. (2024). A shared spatial topography links the functional connectome correlates of cocaine use disorder and dopamine D_{2/3} receptor densities. *Communications Biology*, 7, 1178. DOI: 10.1038/s42003-024-06836-9 [Code]
10. Dong, H.M., Zhang, X.H., **Labache, L.**, Zhang, S., Ooi, L. Q. R., Yeo, B.T. T., Margulies, D. S., Holmes, A. J., & Zuo, X.N. (2024). Ventral attention network connectivity is linked to cortical maturation and cognitive ability in childhood. *Nature Neuroscience*. DOI: 10.1038/s41593-024-01736-x [Code]
9. **Labache, L.**, Petit, L., Joliot, M., & Zago, L. (2024). Atlas for the lateralized visuospatial attention networks (ALANs): insights from fMRI and network analyses. *Imaging Neuroscience*, 2(1–22). DOI: 10.1162/imag_a_00208 [Atlas]

8. **Labache, L.**, Ge, T., Yeo, B.T. T., & Holmes, A. J. (2023). Language network lateralization is reflected throughout the macroscale functional organization of cortex. *Nature Communications*, 14, 3405. DOI: 10.1038/s41467-023-39131-y [Code]
7. Chopra, S., **Labache, L.**, Dhamala, E., Orchard, E. R., & Holmes, A. J. (2023). A Practical Guide for Generating Reproducible and Programmatic Neuroimaging Visualizations. *Aperture Neuro*. DOI: 10.52294/001c.85104 [Web-App] [Code]
6. Forkel, S., **Labache, L.**, Parashkev, N., Thiebaut de Schotten, M., & Hesling, I. (2022). Stroke disconnectome decodes reading networks. *Brain structure & function*, 227(9), 2897-2908. DOI: 10.1007/s00429-022-02575-x
5. Doucet, G. E., **Labache, L.**, Thompson, P. M., Joliot, M., Frangou, S., & Alzheimer's Disease Neuroimaging Initiative. (2021). Atlas55+: Brain Functional Atlas of Resting-State Networks for Late Adulthood. *Cerebral cortex*, 31(3), 1719–1731. DOI: 10.1093/cercor/bhaa321
4. Tzourio-Mazoyer, N., **Labache, L.**, Zago, L., Hesling, I., & Mazoyer, B. (2021). Neural support of manual preference revealed by BOLD variations during right and left finger-tapping in a sample of 287 healthy adults balanced for handedness. *Laterality*, 26(4), 398-420. DOI: 10.1080/1357650X.2020.1862142 [Atlas]
3. **Labache, L.**, Mazoyer, B., Joliot, M., Crivello F., Hesling, H., & Tzourio-Mazoyer, N. (2020). Typical and atypical language brain organization based on intrinsic connectivity and multitask functional asymmetries. *eLife*, 9, e58722. DOI: 10.7554/eLife.58722
2. Hesling, I., **Labache, L.**, Joliot, M., & Tzourio-Mazoyer, N. (2019). Large-Scale plurimodal networks common to listening, production and reading word-lists: an fMRI study combining tasks-induced activation and intrinsic connectivity in 144 right-handers. *Brain structure & function* 224(9), 3075-3094. DOI: 10.1007/s00429-019-01951-4 [Atlas]
1. **Labache, L.**, Joliot, M., Saracco J., Jobard G., Hesling I., Zago L., Mellet E., Petit L., Crivello F., Mazoyer B., & Tzourio-Mazoyer N. (2019). A SENTence Supramodal Areas AtlaS (SENSAAS) based on multiple task-induced activation mapping and graph analysis of intrinsic connectivity in 144 healthy right-handers. *Brain structure & function*, 224(2), 859-882. DOI: 10.1007/s00429-018-1810-2 [Atlas]

Under Review

3. Hesling, I., **Labache, L.** (2026). How Many Handednesses Are There? From Categories to Configurations in Human Laterality. Commentary under review: *Laterality*. Special issue: *How Many Handednesses Are There?*
2. **Labache, L.**, Hesling, I., Zago, L. (2025). Independent Lateralization of Language, Attention, and Numerical Cognition Across Task and Rest. Under Review: *Scientific Reports*. DOI: 10.1101/2025.11.23.690045 [Code]
1. **Labache, L.**, Joliot, M., Doucet, G. E., & Saracco, J. (2020). Study of inter-individual variability of three-dimensional data table: detection of unstable variables and sample. DOI: 10.48550/arXiv.2004.05033 (preprint)

Thesis

Labache, L. (2020). Elaboration Of Brain Network Atlases Underpinning Lateralized Cognitive Functions, Application To The Study Of Inter-individual Variability Of Language. *PhD thesis*, French Alternative Energies and Atomic Energy Commission, **Université de Bordeaux**. NNT: 2020BORD0155 (in French).

Open-source Code & Data

10. Cocuzza, C. V., Chopra, S., Segal, A., **Labache, L.**, Chin, R., Joss, K., Holmes, A. J. (2026). HolmesLab/ClinicalNetDynamics: Clinical Network Dynamics version 1.0.0. *Zenodo*. DOI: 10.5281/zenodo.20074876
9. **Labache, L.** (2025). loiclabache/LUCA_brainAtlas: Lateralized Underpinnings of Comparison and Arithmetic atlas (LUCA) (Labache_2025_LUCA_251130). *Zenodo*. DOI: 10.5281/zenodo.17771073
8. **Labache, L.** (2025). loiclabache/Labache_2025_IndLat: Independent Lateralization of Language, Attention, and Numerical Cognition Across Task and Rest (Labache_2025_IndLat_251130). *Zenodo*. DOI: 10.5281/zenodo.17771776
7. **Labache, L.** (2025). loiclabache/Labache_2025_MolCellLat: The molecular and cellular underpinnings of human brain lateralization (Labache_2025_MolCellLat_250413). *Zenodo*. DOI: 10.5281/zenodo.15207454
6. **Labache, L.** (2024). loiclabache/ALANs_brainAtlas: Atlas for the Lateralized Visuospatial Attention Networks (ALANs) (Labache_2024_ALANs_240214). *Zenodo*. DOI: 10.5281/zenodo.10658842
5. **Labache, L.**, Roger, E., Hamlin, N., Kruse, J., Baciú, M., & Doucet, G. E. (2023). When age tips the balance: a dual mechanism affecting hemispheric specialization for language. *Zenodo*. DOI: 10.5281/zenodo.10253278
4. Chopra, S., **Labache, L.**, Dhamala, E., Orchard, E. R., & Holmes, A. J. (2023). A Practical Guide for Generating Reproducible and Programmatic Neuroimaging Visualizations. *Figshare*. DOI: *Table selector*; 10.6084/m9.figshare.23537316.v1, *R version*; 10.6084/m9.figshare.23537313.v1, *Python version*; 10.6084/m9.figshare.23537319.v1
3. **Labache, L.**, Ge, T., Yeo, B.T. T., & Holmes, A. J. (2023). Language network lateralization is reflected throughout the macroscale functional organization of cortex. *Zenodo*. DOI: 10.5281/zenodo.7869039
2. Mazoyer, B., Tzourio-Mazoyer, N., **Labache, L.**, Zago, L., Hesling I. (2021). BIL&GIN FTT fMRI and handedness. *Dryad*, Dataset. DOI: 10.5061/dryad.cz8w9gj1z
1. Mazoyer, B., Tzourio-Mazoyer, N., **Labache, L.**, Joliot, M., Crivello F. (2020). BIL & GIN sentence and rest asymmetries. *Dryad*, Dataset. DOI: 10.5061/dryad.ht76hdrcf

Presentations

Invited Talks

17. **Behrmann Lab's Meeting**, University of Pittsburgh – Pittsburgh, PA, US. (September 15th, 2026). *The Architecture of Human Hemispheric Specialization: From Functional Networks to Molecular Organization*.
16. **Rutgers Center for Biomedical Informatics & Health Artificial Intelligence Trainee Symposium**, Rutgers University – Piscataway, NJ, US. (July 31st, 2026). *Toward Precision Mapping of Brain Lateralization Across Psychiatric Dimensions*.
15. **Organization for Human Brain Mapping**, Palais 2 l'Atlantique – Bordeaux, FR. (June 15th, 2026). *A Global Perspective on Hemispheric Complementarity Reveals Population-Level Language and Attention Crowding Phenomenon* (Oral Session: Language Across Brain Systems and the Lifespan).
14. **Rising Scholar Seminar**, Rutgers University – Piscataway, NJ, US. (April 13th, 2026). *Multiscale Organization of Human Brain Lateralization: From Molecular Asymmetries to Cognitive Phenotypes*.

13. **Organization for Human Brain Mapping**, Brisbane Convention and Exhibition Centre – Brisbane, AU. (June 26th, 2025). *The molecular and cellular underpinnings of human brain lateralization*. (Oral Session: Language: Anatomy, Networks and Pathology) Youtube Recording
12. **IMA-brain Seminar Series**, Institute of Psychiatry and Neuroscience of Paris – Paris, FR. (January 28th, 2025). *Hemispheric Specialization as a Model for the Emergence of Cognition*.
11. **CAHBIR Seminar Series**, Rutgers University – Piscataway, NJ, US. (November 18th, 2024). *How Our Brains Process Language: Specialization, Individual Differences, Genetics, and Aging*.
10. **Scientific Focus at LaPsyDÉ**, Université Paris Cité – Paris, FR. (September 27th, 2024). *Hemispheric Specialization as a Model for the Emergence of Cognition*.
9. **LPNC Talks series**, Université Grenoble Alpes – Grenoble, FR. (June 1st, 2023). *From the Elaboration of Language Atlases to Understanding Brain Lateralization*.
8. **Ima-Brain**, UCL Institute of Education - London, UK. (April 19th, 2023). *When Age Tips The Balance: a Dual Mechanism Affecting Hemispheric Specialization for Language*.
7. **Psychology's Current Work Series in Neuroscience**, Department of Psychology, Yale University – New Haven, CT, US. (November 11th, 2022). *From the Elaboration of Cognitive Atlases to the Study of Hemispheric Variability*.
6. **Organization for Human Brain Mapping**, Scottish Event Campus – Glasgow, GB. (June 23rd, 2022). *Relationship between Language Lateralization and Global Brain Architecture*. (Oral Session: Events, Memories, and Language) Youtube Recording
5. **Cognitive Neuroscience Meetings**, Department of Psychology, Yale University – New Haven, CT, US. (May 21st, 2021). *From the Identification of Language Network to Typical and Atypical Brain Organization*.
4. **Oslo Virtual Laterality Colloquium**, Department of Psychology, University of Oslo – Oslo, NO. (February 26th, 2021). *Typical and Atypical Brain Organization for Language*.
3. **3rd Doctoral Day of the Frédéric Joliot Institute**, Neursopin, CEA – Paris-Saclay, FR. (June 14th, 2019). *"Elaboration of Brain Atlases of Functional Areas."*
2. **51^{ème} Journées de Statistique**, Université de Lorraine - Nancy, FR. (June 3rd, 2019). *Study of Inter-Individual Variability of Three-Dimensional Data Table: Detection of Unstable Variables and Samples*.
1. **10^e Colloque des Jeunes Chercheurs en Sciences Cognitives**, Ecole Normale Supérieure – Paris, FR. (March 29th, 2019). *SENSAAS, a New Atlas of Language Brain Areas*.

Posters

26. Daube, A., Culhane, B., Joss, K., Cocuzza, C., **Labache, L.**, Baker, J., Yeo, T., & Holmes, A. J. (2026) *A multimodal dataset for the study of brain connectivity and real-world symptom dynamics in affective and psychotic illness*. **Society for Research in Psychopathology**, Hyatt Regency Columbus - Columbus, Ohio, US.
25. **Labache, L.**, Beyh, A., Parkes, L., & Holmes, A. J. (2026) *A Global Perspective on Hemispheric Organization Reveals Population-Level Language-Attention Crowding*. **Organization for Human Brain Mapping**, Palais 2 l'Atlantique - Bordeaux, FR. DOI: 10.5281/zenodo.20817055
24. **Labache, L.**, Hesling, I., Zago, L. (2026) *Language Lateralization Shapes Task, not Rest, in Attentional and Numerical Networks*. **Organization for Human Brain Mapping**, Palais 2 l'Atlantique - Bordeaux, FR. DOI: 10.5281/zenodo.20817055

23. Lawhead, C., Silver, J., Olino, T., **Labache, L.**, Juhng, S., Schwartz, H., Klein, D. (2026) *Longitudinal clustering of psychopathology across childhood and adolescence: an approach toward developmentally-based classification*. **Life History Research Society Meeting**, University of Oslo - Oslo, NO.
22. **Labache, L.**, Chopra, S., Zhang, X.-H., & Holmes, A. J. (2026) *The molecular and cellular underpinnings of human brain lateralization*. **Society of Biological Psychiatry**, New York Marriott Marquis - New York City, US. DOI: 10.1016/j.biopsych.2026.03.434
21. Daube, A., Culhane, B., Joss, K., Cocuzza, C., **Labache, L.**, Baker, J., Yeo, T., & Holmes, A. J. (2026) *A transdiagnostic dataset for mapping functional connectivity and symptom dynamics in affective and psychotic illness*. **Society of Biological Psychiatry**, New York Marriott Marquis - New York City, US. DOI: 10.1016/j.biopsych.2026.03.925
20. **Labache, L.**, Bajwa, W. U., & Holmes, A. J. (2026) *Precision Mapping Reveals Individual Differences in Brain Lateralization Across Psychiatric Dimensions*. **Advancing Rutgers Health AI through Research, Discovery, and Collaboration 2026 Symposium**, College Avenue Student Center - New Brunswick, New Jersey, US.
19. Daube, A., Culhane, B., Joss, K., Cocuzza, C., **Labache, L.**, Baker, J., Yeo, T., Holmes, A. J. (2026) *A transdiagnostic dataset for mapping functional connectivity and symptom dynamics in affective and psychotic illness*. **Society of Biological Psychiatry**, New York Marriott Marquis - New York City, US.
18. **Labache, L.**, Chopra, S., Zhang, X.-H., & Holmes, A. (2025) *The molecular and cellular underpinnings of human brain lateralization*. **Organization for Human Brain Mapping**, Brisbane Convention and Exhibition Centre - Brisbane, AU. DOI: 10.5281/zenodo.15641972
17. Qu, Y. L., Chopra, S., Qu, S., Cocuzza, C. V., **Labache, L.**, Bauer, C. C. C., Morfini, F., Whitfield-Gabrieli, S., Slavich, G. M., Joormann, J., Holmes, A. J. (2025) *Shared and unique lifetime stressor characteristics and brain networks predict adolescent anxiety and depression*. **Society of Biological Psychiatry**, Sheraton Toronto - Toronto, CA.
16. **Labache, L.**, Petit, L., Joliot, M., & Zago, L. (2024) *Atlas for the Lateralized Visuospatial Attention Networks (ALANs): Insights from fMRI and Network Analyses*. **Human Neuroimaging Day**, Centre Broca Nouvelle Aquitaine - Bordeaux, FR.
15. **Labache, L.**, Ge, T., Yeo, B. T. T., & Holmes, A. J. (2023) *Language network lateralization is reflected throughout the macroscale functional organization of cortex*. **Society for the Neurobiology of Language**, Palais du Pharo - Marseille, FR.
14. **Labache, L.***, Roger, E.*, Baciú, M., & Doucet, G. E. (2023) *When Age Tips The Balance: a Dual Mechanism Affecting Hemispheric Specialization for Language*. **Society for the Neurobiology of Language**, Palais du Pharo - Marseille, FR. *authors contributed equally to this work
13. **Labache, L.***, Roger, E.*, Baciú, M., & Doucet, G. E. (2023) *When Age Tips The Balance: a Dual Mechanism Affecting Hemispheric Specialization for Language*. **Organization for Human Brain Mapping**, Palais des congrès de Montréal - Montréal, CA. DOI: 10.13140/RG.2.2.32508.72324 *authors contributed equally to this work
12. Chopra, S., **Labache, L.**, Dhamala, E., Orchard, E., & Holmes, A. J. (2023) *Brain-code: A web-app to generate brain visualisation code templates for R and Python*. **Organization for Human Brain Mapping**, Palais des congrès de Montréal - Montréal, CA.
11. **Labache, L.**, Petit, L., Joliot, M., & Zago, L. (2023) *Visuospatial Attention Networks Evidenced in a Population with Typical Language Brain Organization*. **Organization for Human Brain Mapping**, Palais des congrès de Montréal - Montréal, CA. DOI: 10.13140/RG.2.2.12271.20641

10. **Labache, L.**, Ge, T., Yeo, B.T. T., & Holmes, A. J. (2023) *Language network lateralization is reflected throughout the macroscale functional organization of cortex*. **Society of Biological Psychiatry**, Hilton Bayfront San Diego - San Diego, California, US. DOI: 10.1016/j.biopsych.2023.02.544
9. Ricard, J.A., **Labache, L.**, Chopra, S., Dhamala, E., Jones, G., Harnett, N., Yip, S., & Holmes, A. J. (2023) *The Neural Underpinnings of Cocaine Use Disorder*. **Society of Biological Psychiatry**, Hilton Bayfront San Diego - San Diego, California, US. DOI: 10.1016/j.biopsych.2023.02.639
8. Ricard, J.A., **Labache, L.**, Chopra, S., Dhamala, E., Harnett, N., Jones, G., Yip, S., & Holmes, A. J. (2022) *The network-level correlates of cocaine use disorder*. **Society for Neuroscience**, San Diego, CA, US.
7. Thomas, A., **Labache, L.**, Tsuchida, A., Arsandaux, J., Zago, L., Tzourio, C., Crivello, F., & Samieri, C. (2022) *Dietary pattern and brain structure among young adults*. **Alzheimer's Association International Conference**, San Diego, CA, US. DOI: 10.1002/alz.064049
6. **Labache, L.**, Ge, T., Yeo, B.T. T., & Holmes, A. J. (2022) *Topological Perspective of atypical brain organization*. **Organization for Human Brain Mapping**, Scottish Event Campus - Glasgow, Scotland, GB.
5. **Labache, L.**, Ge, T., Yeo, B.T. T., & Holmes, A. J. (2022) *Atypical language network organization is reflected in the macroscale organization of the cortical sheet*. **Neurobiology of Language: Key Issues and Ways Forward II**, Max Planck Institute for Psycholinguistics – Nijmegen, NL.
4. **Labache, L.**, Joliot, M., Saracco, J., Mazoyer, B., Tzourio-Mazoyer, N. (2018). *"FALCON: a functional atlas of language comprehension networks based on multiple task-induced activation mapping and graph analysis of intrinsic connectivity in 137 healthy right-handers"*. **2nd Doctoral Day of the Frédéric Joliot Institute**, Neursopin, CEA – Paris, FR.
3. Tzourio-Mazoyer, N., Joliot, M., **Labache, L.**, Crivello, F., Zago, L., Hesling, I., Mazoyer, B. (2018). *"Brain language dominance and hand lateralization relationships: insights from the Brain Imaging Lateralization database (BIL&GIN)"*. **North Sea Laterality International Meeting** – Dundee, GB.
2. Hesling, I., **Labache, L.**, Jobard, G., Leroux, G., Tzourio-Mazoyer, N. (2018). *"Heteromodal brain areas commonly activated and asymmetrical in production, listening and reading tasks at the word level: an fMRI study of 144 right-handers from the BIL&GIN."* **North Sea Laterality International Meeting** – Dundee, GB.
1. **Labache, L.**, Tzourio-Mazoyer, N., Jobard, G., Crivello, F., Mazoyer, B., Joliot, M. (2017). *"Tentative atlas of core language areas from fMRI mapping of 6 language tasks in 144 healthy right-handers"*. **Organization for Human Brain Mapping** – Vancouver, CA

Teaching

- 2019 – 2023 Instructor – **University Degree in Neuropsychology**, Université de Bordeaux, Bordeaux – FR, Course title: *Anatomo-functional organization of language*.
- 2018 – 2020 Co-Instructor – **Engineering Degree (M.S.)**, Institut Polytechnique de Bordeaux, Talence – FR, Course title: Statistical modeling and dynamical systems.

Mentoring and Advising

- 2026 – Present Research Assistants, *Sarah Drake (co-supervisor)*
- 2025 – Present Research Assistants, *Alexandra Daube (co-supervisor)*
- 2021 – Present Ph.D. student, *Xi-Han Zhang (co-supervisor)*

- 2021 – Present Ph.D. student, *Yueyue Lydia Qu (co-supervisor)*
- 2023 – 2025 Research Assistants, *Kaley Joss (co-supervisor)*
- 2025 Ph.D. student, *Clément Guichet (co-supervisor)*
- 2024 – 2025 Master-level student, *Kylie Boumbendjé (co-supervisor)*
- 2023 Master-level student, *Léa Auvray (co-supervisor)*
- 2021 – 2023 Research Assistants, *Jocelyn Ricard (co-supervisor)*
- 2021 – 2023 Research Assistants, *Connor Lawhead (co-supervisor)*
- 2018 – 2020 Master-level student, *Martin Constant (co-supervisor)*

Services

Ad Hoc Reviewer: Biological Psychiatry, Brain Structure and Function, Cerebral Cortex, Communications Biology, Human Brain Mapping, Imaging Neuroscience, Journal of Neurophysiology, Journal of Neuroscience, Neurobiology of Language, NeuroImage, Science, Scientific Reports

- 2026 **Grant Reviewer**, *Czech Science Foundation*
- 2022, 2024 **Abstract Reviewer**, *Organization for Human Brain Mapping*
- 2019–2021 **Elected member of the Laboratory Council**, *Institut de Mathématiques de Bordeaux*

Membership

- 2021 – Present **Society for Neuroscience**
- 2021 – Present **Society of Biological Psychiatry**
- 2017 – Present **Organization for Human Brain Mapping**

Referees

References available upon request.

- Referee 1 **Avram J. Holmes** 🏠 – **Associate Professor of Psychiatry**, Holmes Lab – Department of Psychiatry – Rutgers University – Piscataway, NJ, USA
e-mail: avram.holmes@rutgers.edu
- Referee 2 **Monica Baciú** 🏠 – **Professor of Cognitive Neuroscience & Neurologist**, Laboratoire de Psychologie et Neurocognition, CNRS – UMR 5105 – Université Grenoble Alpes – Grenoble, FR
e-mail: monica.baciú@univ-grenoble-alpes.fr
- Referee 3 **Michel Thiebaut de Schotten** 🏠 – **Research Director**, Neurofunctional Imaging Group, CNRS – UMR 5293 – Bordeaux, FR
e-mail: michel.thiebaut@gmail.com